ASAD ABBAS

AI Engineer | Machine Learning & RAG Developer

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with hands-on experience building production-ready intelligent systems spanning Retrieval-Augmented Generation (RAG), Large Language Models, deep learning, and full-stack AI application development. Proven ability to architect scalable backends with FastAPI, integrate LLMs and vector databases into real-world workflows, and deliver end-to-end AI products across healthcare, cybersecurity, education, and productivity domains. Actively pursuing AI/ML internships and freelance opportunities where applied AI creates measurable impact.

TECHNICAL SKILLS

AI & ML: Machine Learning, Deep Learning, RAG Systems, LLMs, NLP, Prompt Engineering, AI Automation

Frameworks & Tools: LangChain, LangGraph, OpenAI API, TensorFlow, PyTorch, Scikit-Learn, FAISS, Vector Databases

Backend: Python, FastAPI, Flask, REST APIs, API Integrations

Frontend: React.js, Next.js, JavaScript, TypeScript, Tailwind CSS

Deployment & Cloud: GitHub, Vercel, Railway, Render, AWS, GCP, Azure

EXPERIENCE

Freelance AI Engineer | *Upwork / Fiverr* *Jan 2025 – Present*

- Design and deploy end-to-end AI applications integrating LLMs, RAG pipelines, and vector databases into production-ready web platforms.
- Build scalable backend APIs using FastAPI and Python to serve ML model inference, real-time data processing, and AI workflow orchestration.
- Develop full-stack AI products with React/Next.js frontends, delivering polished interfaces for healthcare, cybersecurity, and education clients.
- Implement ML/DL models for classification, prediction, and anomaly detection — achieving measurable improvements in client workflows.
- Leverage LangChain and LangGraph to architect multi-step agentic AI workflows and conversational AI systems.

KEY PROJECTS

WanderAI — AI Travel Planner — *FastAPI · LangChain · FAISS · LLaMA 3.3 70B (Groq) · Open-Meteo · React*
Built a worldwide RAG-powered travel planning platform that ingests destination data into a FAISS vector store and uses a 70B LLM to generate personalized, context-aware itineraries with live weather integration. Expanded from a 3-city prototype to full global coverage; deployed with FastAPI backend and a polished responsive frontend.

AI-Powered Cybersecurity Defense System — *Python · ML/DL · Anomaly Detection · FastAPI*
Engineered an AI-driven threat-detection platform that applies anomaly detection and classification models to identify suspicious network behavior and proactively alert on security incidents — reducing manual triage effort for monitored environments.

AI Virtual Voice Assistant — *Python · NLP · Speech Recognition · LLM APIs*
Developed a voice-enabled conversational AI assistant with natural language understanding, intent recognition, and multi-domain task execution — demonstrating end-to-end integration of speech, NLP, and LLM pipelines.

Rice Disease Classification (Computer Vision) — *PyTorch · CNN · Transfer Learning · OpenCV*
Trained a deep learning CNN with transfer learning to detect and classify rice crop diseases from field images, providing automated agricultural decision support with high classification accuracy on a custom annotated dataset.

Phishing URL Detection — *Scikit-Learn · Feature Engineering · Random Forest · Flask*
Built an ML pipeline that extracts lexical and host-based URL features and classifies malicious phishing links in real-time, significantly reducing false-negative rates compared to rule-based baselines.

Heart Disease Risk Prediction — *Python · Scikit-Learn · Pandas · Streamlit*
Developed a supervised ML model for clinical heart disease risk estimation using patient vitals and lab data, packaged as an interactive prediction app for non-technical healthcare users.

EDUCATION

B.S. Information Technology — University of Agriculture, Peshawar *2022 – 2026 (Expected)*

CERTIFICATIONS & LEARNING

- Data Science Bootcamp — Comprehensive training in data analysis, ML algorithms, and model evaluation
- Ongoing: LangChain & LLM application development (self-directed, applied in live projects)